EXPERIMENT – 3 **DATE: 22-10-2026**

I²C-BASED HEALTH MONITORING NODE FOR PULSE AND TEMPERATURE ACQUISITION

Objectives of this Experiment:

1. To understand the basic operation of I²C communication between an Arduino UNO and an LM75AD temperature sensor.
2. To develop an Embedded C program for acquiring temperature data from the LM75AD.
3. To interface a pulse signal source with the digital input of the Arduino UNO.
4. To calculate pulse rate in beats per minute (BPM) from the acquired pulse signal.
5. To display the measured temperature and pulse rate through the Proteus Virtual Terminal.
6. To simulate and verify the health monitoring node using Proteus Design Suite.

OUTPUT PARAMETER TABLE – VALUES TO BE RECORDED

S. No.	Output Parameter	Expected Value / Condition	Practical / Simulated Observation
1	**I²C Device Address**	0x48 detected	Verified
2	**Temperature Acquisition**	Temperature displayed in °C	Verified
3	**Temperature Variation**	Display follows LM75AD setting	Verified
4	**Pulse Input**	Digital pulses received at D8	Verified
5	**Pulse Rate**	Approximately 72 BPM	Verified
6	**Serial Communication**	Data displayed on Virtual Terminal	Verified
7	**Continuous Monitoring**	Temperature and BPM repeatedly displayed	Verified

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Objective	Assessment Criterion	Weightage	Marks
Develop the Proteus circuit	Circuit Design and Configuration	2.5	
Develop the Embedded C program	Program Development	2.5	
Implement sensor processing	Sensor Acquisition and Decision Logic	2.5	
Verify system operation	Simulation and Result Analysis	2.5	
Total		**10**	

Experiment uploaded in GitHub on :

https://github.com/madurivadapalli-tech/192311368.EMBEDDED/blob/main/LAB%20EXPERIMENTS/EXPERIMENT%203.pdf